

Anand Khanna

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PROFESSIONAL SUMMARY

AI/ML Engineer building production **RAG**, **Graph RAG**, **agentic LLM**, and machine learning systems, improving retrieval for **Ray-Ban Meta glasses** at **Meta**. Experienced in **semantic retrieval**, **LLM-as-Judge**, **BERTScore**, **traceability scoring**, and **model evaluation**, with work across **M&A intelligence** and **healthcare automation** using **LangGraph**, **AutoGen**, **Claude**, and **OpenAI/Gemini APIs**. Reduced hallucinations by **40%** and improved analytical accuracy by **35%** through retrieval optimization and AI evaluation.

TECHNICAL SKILLS

AI & LLM Systems

LLMs, RAG Pipelines, Graph RAG, Agentic Workflows, Prompt Engineering, LLM Evaluation, LLM-as-Judge, BERTScore, Semantic Search, Embeddings, Fine-Tuning, Evaluation Dataset Design, FAISS, Vector Databases

Data Engineering & Retrieval

Python (Pandas, NumPy, Matplotlib), SQL (CTEs, Window Functions, Joins), ETL Pipelines, Data Validation & Reconciliation, PostgreSQL, MySQL, MongoDB, Apache Spark

LLM Frameworks & APIs

LangChain, LangGraph, LlamaIndex, AutoGen, Hugging Face Transformers, Claude, OpenAI API, Gemini API

ML & Modeling

PyTorch, TensorFlow, Scikit-Learn, XGBoost, Random Forest, Feature Engineering, Model Training, Hyperparameter Tuning, Cross-Validation, Model Evaluation,

MLOps & Deployment

FastAPI, Flask, REST APIs, Streamlit, Docker, Git, GitHub Actions, CI/CD, MLflow, Model Monitoring, Experiment Tracking, n8n Workflow Orchestration

Systems & Infrastructure

Linux, SSH, Bash/Shell, Device Diagnostics, Log Analysis, Network Troubleshooting, Device Flashing, AWS, Azure, Pinecone, ChromaDB, Weaviate, vLLM, Ollama

EXPERIENCE

• Meta

June 2026 - Present

Senior Test Software Engineer | Python Automation & AI Systems

California, USA

- Enhanced the **RAG system supporting Ray-Ban Meta glasses** and contributed to its **Graph RAG capabilities**, improving semantic retrieval, contextual grounding, and knowledge organization across technical documentation; increased retrieved-context relevance by **25%** across simulated real-world workflows using **Android/iOS-paired and standalone device environments**.
- Designed and deployed **Python-based Claude Skills** for device provisioning, automating serial-number validation, MAC-address updates, host mappings, and synchronized updates across **SERF and LabNet**; reduced an **8-10 step workflow to 2 user actions**, eliminating **90% of manual steps** and saving **10 minutes per phone-device pair** across **2,000+ racks**.
- Developed a **Python-based rack analytics and visualization tool** integrating **SERF inventory data, barcode/QR scanning, and JSON-based host/device mappings**; transformed and validated rack metadata into structured datasets and rendered an **interactive dashboard UI** showing hosts, connected devices, shelf positions, and physical rack locations across **2,000+ racks**.

• Trinity Tree Psychological Services

January 2026 - June 2026

Data Science & Automation Engineer

Remote, USA

- Engineered a production healthcare data and automation platform using **Python, Flask, and Google Sheets API** to validate and manage **1,000+ provider records**, improving workflow reliability and targeting efficiency by **35%** across outreach operations.
- Implemented **caching, rate-limit handling, validation, and data-quality checks** across dashboard and API workflows, reducing external API failures by **90%** while improving operational visibility and response consistency for day-to-day users.
- Automated document-processing workflows using **Python, Gmail API, pdfplumber, regex, python-docx, and Dropbox** to extract data from **200+ HIPAA-sensitive PDFs/month**, reducing manual processing by **50%** and generating standardized reports.

• SowFin

October 2025 - January 2026

AI Engineer

Remote, USA

- Built end-to-end **LLM-powered analytical pipelines** for M&A and investment intelligence using **Claude Sonnet, Exa AI**, and financial APIs, combining chunking, semantic retrieval, and source-grounded generation over **Pitchbook, Crunchbase, and CB Insights** to improve analytical accuracy by **35%**.
- Established a modular **LLM evaluation and benchmarking framework** using **Claude Sonnet as LLM-as-Judge, BERTScore**, traceability scoring, and automated observability checks, improving output consistency and evaluation reproducibility by **30%**.
- Designed and validated a multi-source **Graph RAG knowledge base** linking financial entities and supporting evidence across **Pitchbook, Crunchbase, and CB Insights**, reducing hallucinated outputs by **40%** and increasing evidenced insights by **25%**.

PROJECTS

• Sentiment Analysis using VADER and RoBERTa | Python, PyTorch, Hugging Face, NLP, Streamlit

Fine-tuned **RoBERTa** on Amazon Fine Food Reviews, achieving **92.2% accuracy** and outperforming **VADER (87.5%)**; benchmarked model variants and deployed real-time inference through a **Streamlit** dashboard with side-by-side performance visualization.

• MLOps Maths Score Predictor | Python, Scikit-Learn, MLflow, Docker, Azure, CI/CD

Developed an end-to-end **MLOps pipeline** for student score prediction using **Scikit-Learn**, with **MLflow** for experiment tracking and **Docker** for reproducibility; deployed on **Azure App Services** with CI/CD, improving prediction reliability by **20%**.

• Chat with PDF using Gemini AI | LangChain, FAISS, Gemini, Python, Streamlit

Developed an end-to-end **RAG pipeline** using **LangChain** and **FAISS** for document chunking, semantic retrieval, and context-grounded generation over PDFs; integrated **Gemini AI** and improved contextual response accuracy by **40%** through a **Streamlit** interface.

• Agentic AI – Supply Chain Risk Intelligence Dashboard | FastAPI, Hugging Face Transformers, Streamlit, SQLite, Plotly

Developed a full-stack **agentic AI** platform for supply-chain risk analysis using **FastAPI** and **Streamlit**; integrated **Hugging Face transformer models** for shipment-level risk scoring with a modular backend supporting real-time monitoring, rerouting workflows, and interactive Plotly visualizations.

• Speech Suite - XTTS v2 + Whisper | PyTorch, Whisper, XTTS v2, MarianMT, Gradio

Developed a multilingual speech system integrating **Coqui XTTS v2, OpenAI Whisper**, and **MarianMT** for text-to-speech, speech-to-text, and real-time translation across **13 languages**; deployed an interactive **Gradio** interface for multimodal inference.

EDUCATION

• Arizona State University

Masters in Data Science, Analytics and Engineering (GPA: 3.57/4.00)

August 2023 - May 2025

Tempe, Arizona

• HMRITM, Guru Gobind Singh Indraprastha University

Bachelors in Computer Science Engineering (GPA: 3.67/4.00)

August 2019 - May 2023

New Delhi, India